

Rajesh Sawant

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SUMMARY

Software Engineer with 3+ years of experience building full-stack applications, distributed data pipelines, and AI-powered systems. Currently building scalable LLM-driven extraction pipelines processing 40,000+ websites and multi-architecture AI agent frameworks at Arizona State University. Published researcher in knowledge graph systems at IEEE COMPSAC 2025.

EDUCATION

Arizona State University

Master of Science, Computer Software Engineering

Aug 2024 - May 2026

Tempe, AZ

• **GPA:** 3.76/4

• **Coursework:** Data Structures and Algorithms, Object Oriented Programming, Cloud Computing, Machine Learning, Big Data, Data Mining, Generative AI, Software Security, Testing and Validation, Software Project Management, Distributed Systems

Savitribai Phule Pune University

Bachelor of Engineering, Electronics and Telecommunications

Aug 2016 - Dec 2020

Pune, India

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, SQL, TypeScript, JavaScript, C++, HTML, CSS
- **Frameworks :** React.js, FastAPI, Node.js, Next.js, JUnit, Spring Boot, Tailwind CSS, Jest, Playwright, Redux Toolkit, shadcn/ui
- **AI & Machine Learning :** Scikit-learn, TensorFlow, Pandas, NumPy, RAG, LangChain, LangGraph, Agentic AI, MCP, Prompt Engineering
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, GraphDB, Firestore, Neo4j, Supabase, Pinecone, FAISS
- **Cloud & DevOps:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure, Firebase, GitHub Actions, Docker, Kubernetes
- **Tools & Technologies:** Jira, Confluence, Postman, Mendix, Figma, n8n, Cursor, Claude Code, Protégé, Ollama

WORK EXPERIENCE

Arizona State University

Graduate Service Assistant - Software Developer

Jun 2025 - Present

Tempe, AZ

- Designed and owned scalable backend and frontend workflows for manufacturer onboarding using Next.js, TypeScript, Prisma, FastAPI and MongoDB, improving reliability and reducing manual intervention.
- Built responsive, accessible UI components with shadcn/ui and Tailwind CSS, improving usability across devices and increasing user satisfaction by reducing onboarding time and registration errors.
- Implemented LLM-powered real-time and batch extraction workflows with asynchronous workers, retries, and controlled concurrency, increasing structured data coverage while keeping extraction latency within production limits.
- Built a distributed, queue-driven ETL pipeline processing data from 40,000+ manufacturer websites, using AWS SQS to decouple ingestion, extraction, and reconciliation stages and improve system fault tolerance.

Intelizign Lifecycle Services

Software Engineer

Aug 2021 - Jul 2024

Pune, India

- Designed and delivered enterprise web applications for manufacturing and automotive clients, collaborating with Scrum teams to turn business needs into scalable solutions that improved process efficiency and reduced onboarding time by 30%.
- Developed and managed data pipelines using REST APIs and service-oriented architecture to integrate and unify large datasets across Mendix, Java services, and external systems, increasing operational efficiency by 30% and supporting advanced analytics.
- Built reusable React UI components and contributed to UI/UX design improvements, including conducting UI/UX design workshops and onboarding sessions for new team members to improve development consistency and usability standards.
- Led Agile execution at the team level, contributing to sprint planning, backlog refinement, code reviews, and CI/CD pipelines, while deploying and operating applications on AWS using Docker, Kubernetes to support scalable and reliable production systems.

PROJECTS

KGITP - Intelligent Travel Planning

IEEE

Oct 2024 - Dec 2024

- Led the end-to-end design of a semantic web driven travel planner, applying layered architecture and repository patterns to define system structure and knowledge graph access, and aligning the team on problem formulation.
- Performed ETL across heterogeneous datasets and modeled entities and relationships in a graph database, enabling semantic querying and geospatial filtering.
- Implemented FastAPI endpoints to orchestrate SPARQL queries and ranking logic, and built a React based UI to visualize personalized itineraries and geospatial results.

Sahana - Smart Event and Meetup Platform

Jul 2024 - Present

- Engineered a full-stack event platform with FastAPI and React/TypeScript, implementing Google SSO authentication, JWT-based access control, cursor-based pagination across 12+ API endpoints, and a Redis caching layer with tiered TTLs that reduced redundant Firestore reads and improved query response times for paginated event feeds.
- Built an automated ETL ingestion pipeline triggered via GCP Cloud Scheduler, pulling from the Ticketmaster API across all user cities, deduplicating events through a two-tier strategy (Redis + Firestore fallback), and deploying to Cloud Run with GitHub Actions CI/CD for zero-downtime releases.

- Developed an interest and location-aware friend recommendation engine using weighted scoring (Jaccard similarity on interests, geodistance decay, and attended-category overlap), and built a performance-optimized React frontend with Redux Toolkit state management, Google Maps integration, infinite scroll, and smart cache invalidation on data mutations.